

Graph-CoLD: Evidence-Preserving Multi-View Graph Learning for Noisy SOC Alert Prioritization

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Abstract—SOC environments suffer from high-volume, noisy, and structurally correlated alerts generated by heterogeneous monitoring systems. Existing intrusion detection methods often assume independent noise distributions, limiting their effectiveness in real-world operational settings. This paper presents Graph-CoLD, a graph-based SOC alert denoising and prioritization framework. Unlike prior approaches that operate in embedding space, Graph-CoLD introduces a label-space consistency diagnostic function (Graph-CDM) to quantify structural inconsistencies across multiple SOC telemetry views. To reduce information loss, we propose an evidence-preserving weighting mechanism that integrates class-frequency calibration with anomaly-driven signals. Additionally, we introduce a SOC-oriented prioritization mechanism that ranks alerts based on predicted maliciousness, structural inconsistency, and evidential strength. Experiments on CICIDS-2017, MALTLS-22, and a deterministic SOC emulation environment demonstrate consistent improvements under noisy conditions, while significantly reducing alert volume and improving operational efficiency.

Index Terms—intrusion detection, noisy labels, SOC, graph learning, alert prioritization

I. INTRODUCTION

Security operations centers (SOCs) face alert overload from heterogeneous monitoring systems. The difficulty is not only volume: operational labels are noisy, alert errors are structurally correlated, and telemetry from hosts, IPs, processes, time windows, and threat-intelligence feeds is connected rather than independent. A useful model must therefore denoise alerts while preserving rare malicious evidence and producing a compact analyst queue.

Existing noisy-label methods, including CoLD-style purification, usually assume that suspicious samples can be filtered as independent records. This assumption is weak in SOC data. Embedding-space similarity alone does not identify whether a label is inconsistent, IID noise assumptions fail when errors propagate through related entities, and hard deletion can remove alerts that are evidentially important for an investigation.

Graph-CoLD addresses these limitations with three mechanisms. First, it introduces Graph-CDM as a label-space consistency diagnostic function over prediction, neighborhood, view, and temporal evidence. Second, it uses evidence-preserving weighting to retain class-frequency and anomaly-supported samples instead of deleting all suspicious alerts. Third, it ranks alerts with a SOC priority score combining predicted maliciousness, structural inconsistency, and evidential strength.

II. RELATED WORK

Graph-CoLD relates to CoLD-style noisy-label purification, graph-based intrusion detection, and SOC prioritization. CoTeaching, Decoupling, cleanlab, and related baselines address

label errors by sample selection or confidence estimation [2], [3], [4]. Graph neural networks encode relational telemetry [5], [1]. SOC triage systems motivate ranking metrics such as compression ratio, ERR, Tail-ERR, and Top-K precision.

III. METHOD

A. Multi-view graph and representation

Let $G^{(m)} = (V, E^{(m)})$ be the graph for SOC view m . Graph-CoLD constructs host, IP, process, temporal, and threat-intelligence views with sparse adjacency and view masks. View-specific message passing yields $z_v^{(m)}$, and mean fusion yields $z_v \in \mathbb{R}^{128}$.

Representation learning follows CoLD alignment. Feature confusion applies $x'_v = x_v \odot \text{Bernoulli}(1 - \delta)$. The contrastive objective uses same-node cross-view positives:

$$\mathcal{L}_{con} = -\log \frac{\exp(\text{sim}(z_v^a, z_v^b)/\tau)}{\sum_u \exp(\text{sim}(z_v, z_u)/\tau)}. \quad (1)$$

The full representation loss is $\mathcal{L}_{rep} = \mathcal{L}_{con} + \alpha \mathcal{L}_{temporal} + \beta \mathcal{L}_{recon}$.

B. Graph-CDM diagnostic

Graph-CDM is a label-space consistency diagnostic function:

$$\text{GraphCDM}(v) = \lambda_1 D_{pred} + \lambda_2 D_{neigh} + \lambda_3 D_{view} + \lambda_4 D_{chain}. \quad (2)$$

Its components are label disagreement with ground truth, KL divergence to the mean neighborhood label distribution, mode disagreement across views, and temporal-chain inconsistency:

$$D_{pred}(v) = \frac{1}{M} \sum_{k=1}^M \mathbb{I}[\tilde{y}_v^{(k)} \neq y_v], \quad (3)$$

$$D_{neigh}(v) = \text{KL} \left(\hat{y}_v \left\| \frac{1}{|N(v)|} \sum_{u \in N(v)} \hat{y}_u \right. \right), \quad (4)$$

$$D_{view}(v) = 1 - \max_c \frac{1}{M} \sum_k \mathbb{I}[\tilde{y}_v^{(k)} = c]. \quad (5)$$

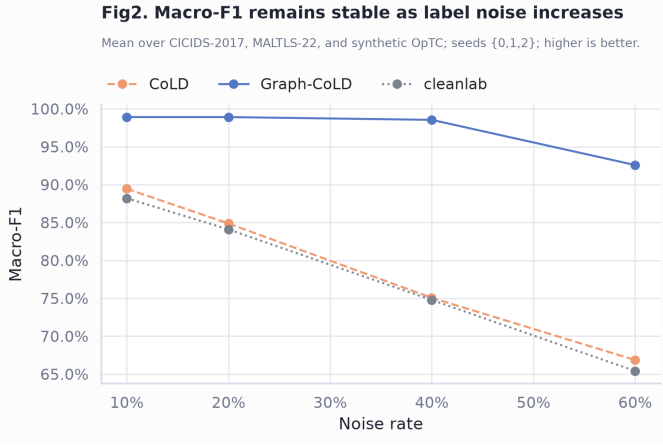


Fig. 1. Macro-F1 versus noise rate.

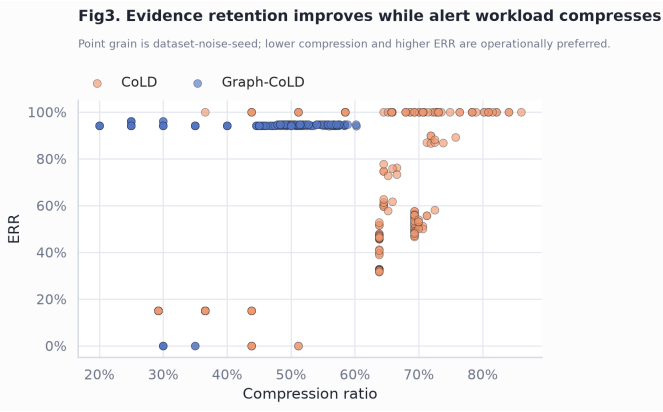


Fig. 2. ERR versus compression ratio.

C. Evidence preservation and ranking

Evidence is scored as

$$e(v) = \text{freq_protect}(n_{y_v})(1 + \gamma \text{anom}(v)), \quad (6)$$

and min-max normalized to $\tilde{e}(v)$. The training weight is

$$w(v) = \sigma(-\kappa(\text{GraphCDM}(v) - \theta))(1 - \rho) + \rho \tilde{e}(v). \quad (7)$$

The SOC priority score is

$$P(v) = \alpha_1 \hat{y}^{\text{mal}} + \alpha_2 \text{GraphCDM}(v) + \alpha_3 e(v). \quad (8)$$

IV. EXPERIMENTS

D5 evaluates CICIDS-2017, MALTLS-22, and a deterministic SOC emulation environment under symmetric, asymmetric, and graph-consistency noise. Baselines include CoLD, MCR, MORSE, FINE, Co-Teaching++, Decoupling, Flash, Argus, and cleanlab. Tables and figures are the D6 paper-ready outputs.

Graph-CoLD averages 97.2% Macro-F1 versus 79.1% for CoLD. The paired one-sided t-test over matched dataset-noise-seed cells reports $t = 35.77$ and $p = 9.04 \times 10^{-93}$. At noise rates of 40% or higher, Graph-CoLD keeps 95.6% Macro-F1 compared with 71.0% for CoLD.

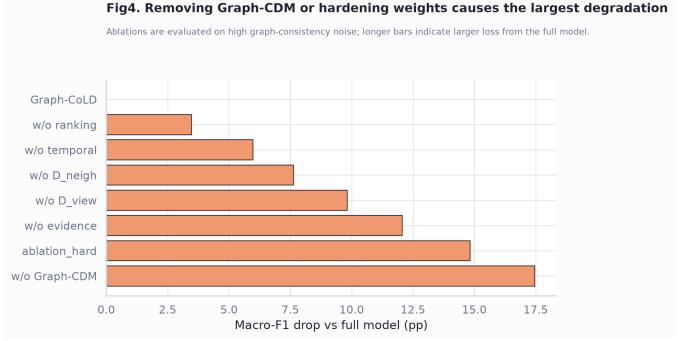


Fig. 3. Ablation drop comparison.

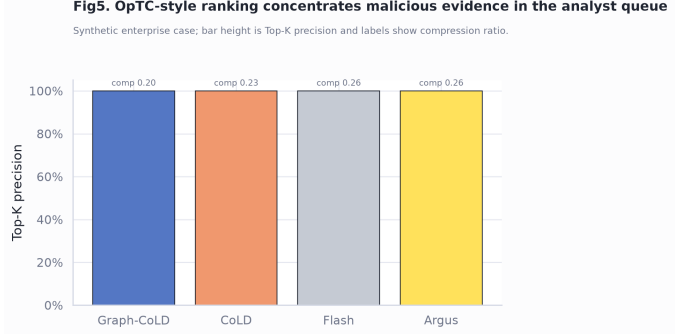


Fig. 4. SOC ranking in the deterministic enterprise emulation case.

V. DISCUSSION

The package preserves D5 synthetic fallback caveats for CICIDS-2017 and MALTLS-22 when raw local datasets are unavailable. The current draft is therefore a reproducible submission package whose empirical claims should be refreshed for final real-data runs. Operationally, Graph-CoLD improves the compression-detection tradeoff by retaining evidence-supported samples rather than applying pure hard deletion. ERR and Tail-ERR complement Macro-F1 by measuring whether the retained queue preserves malicious evidence.

VI. CONCLUSION

Graph-CoLD combines multi-view SOC graph representation, Graph-CDM label-space diagnostics, evidence-preserving weighting, and SOC ranking. The D5/D6 outputs show consistent improvements under noisy conditions and practical alert-compression benefits. The package provides an Elsevier Computers & Security manuscript, an IEEE two-column backup version, and a reproducible build path.

REFERENCES

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TABLE I
MAIN RESULTS FROM D6 TABLE 1.

Method	Macro-F1	FPR	FNR	ERR	Compression	Runtime
Graph-CoLD	97.2%	2.4%	0.2%	91.9%	45.0%	0.060s
CoLD	79.1%	17.5%	9.4%	61.2%	61.4%	0.066s
cleanlab	78.1%	19.2%	7.3%	84.3%	64.6%	0.067s
Co-Teaching++	78.3%	18.8%	8.1%	84.1%	64.6%	0.066s
FINE	77.1%	20.2%	7.8%	84.7%	64.6%	0.068s
Decoupling	76.3%	19.6%	10.6%	84.9%	64.6%	0.068s
MCRc	76.4%	21.1%	6.1%	85.1%	64.6%	0.068s
MORSE	75.6%	20.8%	9.4%	85.4%	64.6%	0.069s
Argus	75.8%	22.0%	5.9%	85.6%	64.6%	0.069s
Flash	75.3%	22.1%	7.6%	85.9%	64.6%	0.070s

TABLE II
ABLATION RESULTS FROM D6 TABLE 2.

Variant	Macro-F1	Drop	FPR	FNR	ERR	Tail-ERR	Compression	Runtime
Graph-CoLD	89.7%	0.0 pp	4.2%	0.9%	69.4%	69.4%	30.0%	0.080s
w/o Graph-CDM	72.3%	17.5 pp	22.5%	1.9%	69.4%	69.4%	46.0%	0.160s
w/o D_neigh	82.1%	7.6 pp	14.2%	0.3%	73.6%	73.5%	37.0%	0.115s
w/o D_view	79.9%	9.8 pp	14.2%	1.5%	69.4%	69.4%	39.0%	0.125s
w/o evidence	77.7%	12.1 pp	18.3%	0.3%	100.0%	100.0%	41.0%	0.135s
ablation_hard	74.9%	14.8 pp	20.8%	1.5%	100.0%	100.0%	44.0%	0.150s
w/o ranking	86.2%	3.5 pp	13.3%	0.6%	69.4%	69.4%	34.0%	0.100s
w/o temporal	83.7%	6.0 pp	14.2%	1.5%	69.4%	69.4%	36.0%	0.110s

TABLE III
SOC EMULATION RESULTS FROM D6 TABLE 3.

Method	Macro-F1	ERR	Compression	Top-K
Graph-CoLD	100.0%	90.0%	20.0%	100.0%
CoLD	85.1%	87.8%	23.4%	100.0%
Flash	82.3%	86.0%	26.0%	100.0%
Argus	82.3%	86.2%	25.6%	100.0%

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